Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.

Team Members	Role/ Responsibilty	Email
Saddam Adam	Bayesian probability	s.adam@alustudent.com
Rolande Tumugane	Gradient descent manual calculations	r.tumugane@alustudent.com
IGIHOZO Nyituriki Marie Colombe	Gradient descent Python implementation	c.nyituriki@alustudent.com
Milena Rwakunda	README documentation, EM algorithm implementation, repository management	m.rwakunda@alustudent.com

Task Allocation and Tracker

	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Milena implemented the EM algorithm from scratch, covering the E-step, the M-step, and the tracking table for iterations zero, one, and two.	Milena	Sat Jun 27	Completed	Yes	
	Milena wrote the README that the team used to guide the presentation.	Milena	Mon Jun 29	Completed	Yes	
	Milena set up and managed the shared GitHub repository so every commit is logged.	Milena	Wed Jul 1	Completed	Yes	
	Saddam loaded the IMDb dataset and selected the positive and negative keywords for the sentiment analysis.	Saddam	Thu Jul 2	Completed	Yes	
	Saddam implemented Bayes theorem in plain Python to compute the prior, likelihood, marginal, and posterior for each keyword.	Saddam	Thu Jul 2	Completed	Yes	
	Saddam put the keyword probabilities into clean tables for the presentation.	Saddam	Thu Jul 2	Completed	Yes	
	Rolande worked through the manual gradient descent calculations by hand, showing the chain rule for both derivatives.	Rolande	Fri Jul 3	Completed	Yes	
	Rolande combined all the handwritten calculations into one neat PDF for submission.	Rolande	Fri Jul 3	Completed	Yes	
	Colombe built the SciPy function that computes the derivative of the linear equation.	Colombe	Sat Jul 4	Completed	Yes	
	Colombe wrote the update loop that adjusts m and b over several iterations, keeping every step visible.	Colombe	Sat Jul 4	Completed	Yes	
	Colombe built the two Matplotlib plots showing how m and b change and how the error drops over each iteration.	Colombe	Sat Jul 4	Completed	Yes	
	The team booked the presentation slot and confirmed every member is added to the invite.	All	Sun Jul 5	Completed	Yes	
	The team put together the final Jupyter notebook with every part and ran through the presentation together.	All	Sun Jul 5	Completed	Yes	

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
Fri Jun 26	Kickoff meeting to split the four parts of the assignment among the team.	Saddam	Saddam, Rolande, Colombe, Milena	Agreed on who would lead each part and picked the keywords for the Bayesian section.	Everyone starts on their assigned part.	

Wed Jul 1	Progress check on the EM algorithm and the Bayesian probability sections.	Saddam	Saddam, Rolande, Colombe, Milena	Compared everyone's free time and picked a slot that worked for the whole team.	Book the slot and add everyone to the calendar invite.	
Fri Jul 3	Progress check on the gradient descent code and the plots. Saddam Saddam, Rolande, Colombe, Milena Looked over the SciPy function, the update loop, and the two plots. Start pulling everything into one notebook.	Rolande	Saddam, Rolande, Colombe, Milena	Reviewed the tracking table, the live demo, and the keyword probability tables.	Finish the remaining code and start on the visualizations.	
Sat Jul 4	Meeting to choose and book our presentation slot.	Milena	Saddam, Rolande, Colombe, Milena	Looked over the SciPy function, the update loop, and the two plots.	Start pulling everything into one notebook.	
Sun Jul 5	Final run-through before the deadline.	Colombe	Saddam, Rolande, Colombe, Milena	Combined the notebook, the README, and the calculation PDF, then rehearsed the presentation.	Submit the assignment and confirm the presentation booking.	